MIL-P-50890(MU)
30 August 1971
Superseding
See 6.4

MILITARY SPECIFICATION

PROJECTILE, TRAINING, 8 INCH, XM440 SHIPPING ASSEMBLY

1. SCOPE

1.1 This specification covers requirements, quality assurance provisons, and the preparation for delivery criteria for one type of projectile, training, 8 inch, XM440 shipping assembly.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

TT-C-490 - Cleaning Method and Pretreatment of Ferrous Surfaces for Organic Coatings

MILITARY

MIL-C-50565(MU) - Container Shipping and Storage, Atomic Projectile: M500

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-1167 - Ammunition Data Cards
MIL-STD-1168 - Lot Numbering of Ammunition

DRAWINGS

ORDNANCE CORPS

8786495 - Ogive

FSC 1110

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8787846 - Plug Nose
8844546 - Body Assembly
8844547 - Base
8844550 - Body
8848823 - Container, Metal, Ammo for Proj Atomic, 8 Inch, Tng, XM440, Shipping Assembly Pkg 1 of 2
8860176 - Projectile, Training, 8 Inch, XM440 Shipping Assembly
ET-8860176 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The projectile shall comply with all requirements specified on the drawing (dwg) 8860176 with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 Paint adhesion. Paint adhesion shall comply with Paint Adhesion, Section 3, TT-C-490 when tested as specified in 4.4.1.

3.4 Workmanship.

3.4.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.4.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.4.3 Burrs. Parts shall be free from burrs.

3.4.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.4.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise speci-

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fied in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Product acceptance inspection.

4.3.1 Lot formation. Lot formation shall be in accordance with MIL-STD-1168. Changes in lot numbers shall be recorded on Ammunition Data Cards in accordance with MIL-STD-1167 and MIL-STD-1168.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

Table I
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

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Table II
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-9	all	14-16	12	28-33	16	77-121	20
10	9	17-18	13	34-41	17	122-273	21
11	10	19-22	14	42-54	18	over 273	22
12-13	11	23-27	15	55-76	19		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Body (see dwg 8844550 covering a detail of dwg 8844546).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Flange diameter (7.990)	Gage
102.	Chamfer present (.12)	Visual
103.	Left inside diameter (2.95)	Gage
104.	Pitch diameter of inside thread	Gage
105.	Minor diameter of inside thread	Gage
106.	True position of inside thread	Gage
107.	Distance from bottom of inside thread clearance to shoulder of flange (.155)	Gage
108.	Distance from bottom of inside thread to shoulder of flange (.40)	Gage
109.	Distance from top of thread to shoulder of flange (1.20)	Gage
110.	Diameter of inside thread clearance, min (2.93)	Gage
111.	True position of right cavity diameter	Gage
112.	Diameter of right cavity (2.82)	Gage
113.	Depth of right cavity min (11.73)	Gage
114.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8844550)	Visual
Minor:		
201	Workmanship (see 3.4)	Visual-Manual

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4.3.2.1.2 Body assembly (see dwg 8844546) covering a detail of dwg 8860176, before painting.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Contour of rotating band	Visual
102.	Diameter between small grooves rotating band (8.155)	Gage
103.	Outside diameter (7.990)	Gage
104.	Diameter of right face (7.89)	Gage
105.	Total length	Gage
106.	No visable gap in joints as noted	Visual
107.	Objective evidence (see 6.2) for protective finish as noted in accordance with dwg 8844546	Visual
108.	Pin holes filled flush	Visual
109.	Components properly assembled	Visual
Minor:		
201.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.3 Plug nose (see dwg 8787846) covering a detail of dwg 8860176.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Distance to outside datum diameter (.660)	Gage
102.	Distance from shoulder to end (2.27)	Gage
103.	Objective evidence (see 6.2) for protective finish in accordance with dwg 8787846	Visual
Minor:		
201.	Surface finish	Visual
202.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.4 Ogive (see dwg 8786495) covering a detail of dwg 8860176.

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:
101. Distance to left outside datum diameter (2.650).....Gage
102. Distance to right outside datum diameter (8.215).....Gage
103. Objective evidence (see 6.2) for protective finish in accordance with dwg 8786495.....Visual

Minor:
201. Surface finish.....Visual
202. Workmanship (see 3.4).....Visual/Manual

4.3.2.1.5 Base (see dwg 8744547) covering a detail of dwg 8860176.

Categories	Characteristics	Method of Inspection

Critical: None defined

Major:
101. Flange diameter (7.97).....Gage
102. Hub diameter (3.96).....Gage
103. Diameter of left tapered inside thread clearance min (3.53).....Gage
104. Pitch diameter of left inside thread.....Gage
105. Minor diameter of left inside thread.....Gage
106. Diameter of 2 hole in flange (.375).....Gage
107. True position of 2 holes in flange.....Gage
108. Pitch diameter of right inside thread.....Gage
109. Minor diameter of right inside thread.....Gage
110. Inside diameter adjacent to right inside thread (3.05).....Gage
111. Width of left inside thread clearance, max (.24).....Gage
112. Distance from left face to right side of flange (1.25).....Gage
113. Total length (2.62).....Gage
114. Depth of inside slots (1.85).....Gage
115. Depth of left inside thread including thread clearance (1.33).....Gage
116. Depth of hub (.28).....Gage
117. Depth of tapered inside thread clearance (.15).....Gage
118. Diameter of inside slots (3.35).....Gage
119. Width of inside slots (.720).....Gage
120. Location of inside slots.....Gage
121. Objective evidence (see 6.2) for protective finish in accordance with dwg 8844547.....Visual

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Minor:
201. Surface finish.....Visual
202. Workmanship (see 3.4).....Visual/Manual

4.3.2.1.6 Projectile, training, 8 inch, M440 Shipping assembly (see dwg 8860176).

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major		
101.	No components missing or improperly assembled	Visual
Minor:		
201.	Painting	Visual
202.	Marking	Visual
203.	Workmanship (see 3.4)	Visual/Manual

4.3.2.1.7 Container, metal, Ammo, for projectile atomic, 8 inch, Tgn, XM440, Shipping assembly Pkg 1 of 2 (see dwg 8848823.

Categories	Characteristics	Method of Inspection
Critical: None defined.		
Major:		
101.	Objective evidence (see 6.2) for container in accordance with MIL-C-50565(MU)	Visual
102.	Container properly packed, no parts missing	Visual
103.	Contents tight in container	Manual
104.	Torque on cover screws	Gage
105.	Seals properly assembled	Visual
106.	Marking	Visual
Minor:		
201.	Workmanship (see 3.4)	Visual/Manual

4.3.3 Testing. The test shall be performed.

Test	Test Classification
a. Paint adhesion	Minor

This test is not destructive, samples tested may be returned to the lot.

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4.3.3.1 Paint adhesion (see 3.3). The projectile shall be tested in accordance with 4.4.1. If any item fails to comply with 3.3, the item shall be rejected and removed from the lot.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). ELs for assemblies are listed on the Equipment Tabulation, ET-8860176. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accurancy tolerances of the test equipment used.

4.4.1 Paint adhesion (see 3.3). Paint adhesion shall be tested in accordance with Paint adhesion, section 4, TT-C-440.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The projectile shall be packaged in accordance with dwg 8848823 using the specified moisture barrier.

5.1.2 Level C. The projectile shall be packaged in accordance with dwg 8848823 except that the moisture barrier may be omitted.

5.2 Packing.

5.2.1 Level A and C. The projectile shall be packed in accordance with dwg 8848823.

5.3 Marking. Marking shall be as specified in accordance with dwg 8848823.

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5.4 Data cards (see 6.1.2). Data cards shall be prepared in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify:
6.1.1 Procurement requirements.

a. Title, number, and date of this specification.
b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1 and 5.4). Data specified in 4.3.4.1 and 5.4 will be delivered as identified on a numbered DD Form 1664 when specified on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commanding Officer, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SMUPA-NR4.

6.4 Supersession data. This specification includes the requirements of Purchase Description PA-PD-2015 (Rev B) dated 31 May 1966.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No.1110-A023

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